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Digital twin-enabled anti-collision system for smart construction sites

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Highlights:

- Anti-collision system is developed to improve safety for HRIs.
- DT system with "perception, analysis, decision-making, execution" is proposed.
- Multi-sensor fusion approach is designed for comprehensive on-site perception.

Abstract: The construction industry is one of the most perilous sectors globally, with significantly higher fatality rates compared to other industries. Collision with heavy construction machines is a major cause of worker injuries and fatalities. This research proposes a digital twin-enabled anti-collision system to provide collision warnings and improve construction safety. A five-dimensional digital twin model encompassing the physical layer, digital data layer, virtual entity layer, service layer, and connection layer is established. The physical layer designs multi-sensor fusion approach to respectively collect image data, depth data, and motion data, achieving comprehensive perception of construction sites. The digital data layer designs multi-modal analysis methods to obtain on-site information such as category, position and motion status. The virtual entity layer integrates geometry model, physical model, behavior model, and rule model to achieve the synchronization between physical entities and digital entities. It also divides construction sites into dangerous zones, warning zones, and safe zones to perform collision warnings based on the status of construction robots and workers. The services layer provides supervision, motion states monitoring, zone prediction, and data management for the DT system. The connections layer establishes the information interconnection and synchronization between these layers. Case study is conducted to verify its performance in visualization, motion states monitoring and collision warning. It is proven that the proposed method can achieve a prediction accuracy of over 96% for dangerous zones and over 95% for warning zones. The proposed digital twin-enabled anti-collision system can greatly improve the safety for construction sites.

Keywords: digital twin; collision warning; smart construction

1. Introduction

The construction industry is one of the most dangerous sectors due to its high rates of accidents and fatalities [1]. As statistics from the Hong Kong government in 2023, the construction industry has the

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highest accident and fatalities rates among all industries. Its accident rate per 1000 workers is 29.1, double that of the manufacturing industry. Its fatality rate per 1000 workers is 0.162, which is three times as much as the manufacturing industry.

Automation and robotics are considered alternatives to alleviate safety concerns in construction [2]. McKinsey estimates that more than 46% of construction machines will be replaced by robots by 2025 [3]. Although construction robots can relieve workers from hazardous tasks, collisions between human-robot interactions (HRIs) are still inevitable. Hence, it is still necessary to design an anti-collision system to improve construction safety [4], as shown in Figure 1. Especially in Hong Kong, construction machines and workers need to operate in narrow spaces, which significantly increases the likelihood of collisions.

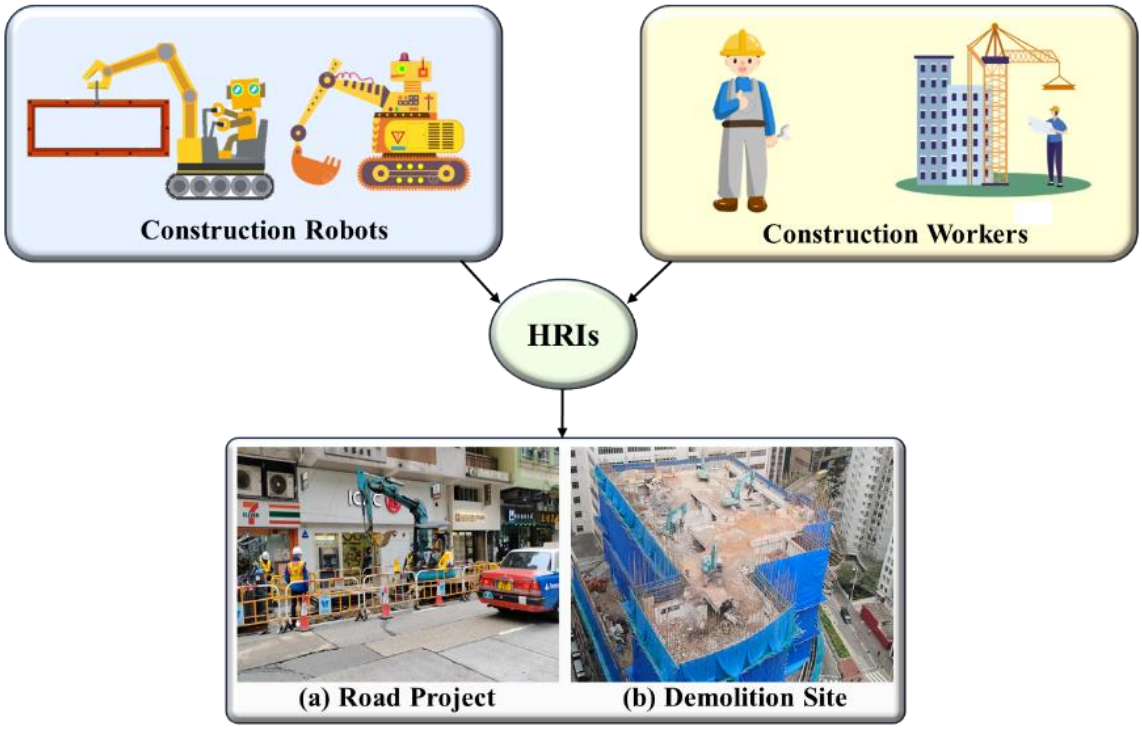


Figure 1. Human robots interactions in construction sites.

The safety monitoring methods for HRIs have been extensively studied in various industries such as manufacturing, construction, and agriculture to improve their efficiency and safety [5,6]. Multi-source sensors [7] such as cameras, Bluetooth devices, Inertial Measurement Units (IMUs), and Ultra-wideband (UWB) sensors can be used for data collection in HRIs. RGB-D cameras and UWB sensors can be employed for proximity estimation to prevent potential accidents between heavy machines and workers. Machine learning methods [8] have also been discussed for the processing, fusion, and prediction of multi-source data. Computer vision methods [9] have been studied for the location, pose estimation, and activity recognition of robots and humans. Intelligent decision-making methods such as space partitioning and bounding volume hierarchy have also been studied for collision detection of crane-lifts in modular integrated construction [10]. Besides, digital systems [11] that combine various sensors have also been designed for proximity estimation, activity recognition, and collision warnings in HRIs. Based on predicted collision events, these systems will take corresponding actions (e.g., giving warnings to humans, stopping robots, or modifying robots' paths) to ensure safety.

Although significant achievements have been made, the safety monitoring methods for HRIs in complicated and dynamic construction sites remain challenging:

HRIs with more uncertainties: Compared with the manufacturing industry, HRIs in construction sites are characterized by significant uncertainty [12]. In manufacturing workshops, robots are installed in standard production lines, where they perform operations efficiently with fixed HRI modes. However, in some construction projects (e.g., demolition projects, road projects), robots need to move proactively, adjust their postures dynamically, and carry out operations until the projects are completed. Therefore, HRIs in construction sites do not follow fixed modes and involve more uncertainties. The comprehensive perception of on-site robots and workers is the basis for safety monitoring of HRIs in construction sites.

High risks of collision: Collision risks in construction are particularly prominent, and collisions with construction robots—often heavy machines—are major causes of fatalities [13]. For instance, statistics show that in Germany last year, collision-related incidents accounted for 36% of all fatal accidents on construction sites. On a global scale, collision-related accidents make up 22% of all construction accidents, solidifying collisions as a major contributor to on-site incidents. In road construction projects specifically, over 50% of fatal injuries to road workers result from collisions with motor vehicles [14]. Therefore, precise anti-collision methods are considered key to improving the safety of HRIs with high risks.

Complex HRIs system: Although robots can greatly improve the level of automation in construction, they also introduce new challenges to HRIs. The occlusion and noise generated by robots can affect workers' perception of the environment, potentially leading to safety hazards. Additionally, there is information that workers are unable to perceive, such as the planned motion paths of robots and their next actions. As a result, HRI systems are characterized as complex systems that involve the synchronization of heterogeneous data, the integration of analysis algorithms, information communication between different roles, seamless decision-making, and services for various stakeholders.

In conclusion, the main objectives of this research are as follows:

(1) To develop multi-sensor perception and analysis methods which can realize comprehensive perception of machines, workers, and construction sites.

(2) To design on-site collision prediction and warning methods for intelligent decision-making, so as to improve the safety of HRIs in construction sites.

(3) To propose an efficient HRIs system which can realize the seamless integration of the "perception, analysis, decision-making, execution".

As a rapidly evolving technology, the digital twin (DT) [15] is widely regarded as one of the most promising technologies for achieving real-time information synchronization in dynamic and complex scenarios. DT refers to a virtual model that provides a comprehensive description of a physical entity, enabling the seamless integration of virtual space with physical space. Multi-source data collected from a physical entity is analyzed and mapped to the virtual entity for simulation, monitoring, and maintenance [16]. DT can also be applied in the safety monitoring system of HRIs, enabling real-time perception and analysis between physical and virtual construction sites, as well as timely decision-making and monitoring.

To achieve these objectives, this paper combines artificial intelligence (AI) technologies and digitalization techniques to develop a digital twin-enabled anti-collision system. Firstly, a multi-sensor fusion approach is proposed to realize a comprehensive understanding of entities in physical construction sites. Besides, a multi-modal analysis model is developed to enable real-time and comprehensive monitoring of HRIs in smart construction sites. Moreover, a 5-Dimensional DT system is designed to

achieve the closed-loop optimization of "perception, analysis, decision-making, execution" for HRIs in construction sites.

The paper is organized as follows: Section 2 provides a comprehensive review of collision warning methods for HRI and digital twin in smart construction. Section 3 systematically describes the proposed digital twin-enabled anti-collision system. Section 4 presents a series of experiments conducted to evaluate the proposed methods and system. Finally, Section 5 provides a summary of the conclusions and future work.

2. Literature review

This section provides a comprehensive review of recent research works, focusing on collision warning methods for HRI, digital twin in smart construction.

2.1. Collision warning methods for HRI

Collision early warning in HRI primarily comprises three interconnected components: perception, analysis, and decision-making, each playing a critical role in ensuring operational safety.

Perception serves as the foundational step, involving the collection of environment and object-related information through diverse sensing technologies. Ultra-Wideband (UWB) and Global Positioning System (GPS) [17] are commonly deployed to obtain real-time location data of objects. Cameras capture visual image data, while thermal sensors acquire temperature-enhanced images to distinguish humans from inanimate objects [18]. Additionally, RGB-D cameras and LiDARs provide 3D point cloud data, enabling detailed spatial reconstruction [19]. Inertial Measurement Units (IMUs) are mounted on objects to collect motion parameters such as velocity, acceleration, and angular rates [20]. Wearable devices have also been extensively studied for collecting both physical and mental data of workers [21].

Analysis focuses on interpreting the collected data to comprehensively understand the motion states of interacting entities. This step relies on advanced algorithms, including deep learning and computer vision techniques [22], to extract meaningful features. For instance, Convolutional Neural Networks (CNNs) are designed to classify, detect, and segment workers and robots from images or videos [23]. Temporal features are often integrated into Long Short-Term Memory (LSTM) network to analyze motion trajectories and patterns, facilitating activity recognition and continuous trajectory monitoring [24]. Moreover, 3D CNN-based methods can detect proximity between robots and workers, a key indicator for triggering early collision warnings [25].

Decision-making involves risk identification and the formulation of safety actions based on analyzed data and predefined rules. Core tasks include collision warning activation, path planning, and trajectory adjustment. For example, an integrated octree and bounding box strategy has been applied to collision detection in crane-lift path planning, enabling efficient spatial conflict checks [26]. In dynamic construction sites, occupied areas of workers and hazardous objects can be mapped onto a lane map, with collision warnings triggered by monitoring proximity thresholds. Collision warning rules have also been studied as varied scenario and application requirements, including static distance (SD) rules, time-to-collision (TTC) rules, spatial occupancy (SO) rules. The SD rules are a common approach in industrial settings where a fixed safety threshold (e.g., 2 meters) is defined. Warnings are given when the distance between a robot and a worker falls below this threshold [27]. The TTC rules are used in

high-speed environments (e.g., autonomous forklifts). TTC calculates the time until a potential collision based on relative velocity and distance, with warnings adjusted dynamically (e.g., an alert is triggered when TTC < 3 seconds) [28]. The SO rules are also used in confined spaces, such as construction tunnels. Here, warnings are activated when a worker enters a predefined "dangerous zone" around a robot regardless of distance, so as to account for limited working space [29]. Furthermore, uncertainty factors (e.g., sensor noise, sudden human movements) are increasingly incorporated into decision-making models to enhance accuracy [30]. Knowledge graph methods have also been widely adopted for risk identification and safety management, enabling structured reasoning about complex interaction scenarios [31]. This multi-layered framework highlights the importance of aligning perception accuracy, analytical depth, and context-specific warning rules to address the dynamic and uncertain nature of HRI environments.

2.2. Digital twin in smart construction

With the rapid development of information technology, digital systems [32,33] such as cyber-physical system (CPS), edge-cloud system, and DT system have been widely applied in the construction industry.

Cyber-physical synchronization between the ongoing construction progress and building information models (BIM) can be realized via the Internet of Things (IoT), cloud computing, and 5G technology, allowing stakeholders to share on-site information and make more efficient decisions [34]. Besides, multi-dimensional information can also be integrated with BIM platforms for various tasks such as scheduling, planning, cost estimation, and daily storage. This provides more comprehensive and extensive information synchronization and integration for construction projects [35]. Liu *et al.* [36] developed a CPS for real-time monitoring and visualization of greenhouse gas emissions in prefabricated construction. The operating states of cranes, transfer vehicles, and construction machines are monitored using acceleration sensors, GPS, and Wi-Fi to calculate greenhouse gas emissions. Machine model numbers, quantities, power, and greenhouse gas emission factors are stored in the cyber database. Furthermore, workers can also be incorporated to design human-cyber-physical systems (HCPS) [37], aiming to improve workers safety and well-being.

DT systems [38] are digital representations of physical systems that possess the same properties and synchronized information as real-world entities. DT systems can be used for full lifecycle operations, simulations, and management, ultimately enabling better decision making. The applications of DT in construction have also been widely studied. Five-dimensional digital twin models are utilized for closed-loop control of the "hanging-adjusting angles-fine positioning-piling" process to avoid collisions [39]. The biggest advantage of DT is its ability to integrate with various technologies including AI, cloud computing, machine learning, and communication technologies to realize more accurate status monitoring and information synchronization. Facilitated by these advanced technologies, physical entities can be coordinated with virtual entities to improve the precision of automatic assembly of bricks and components with the collaboration of DT systems [40]. In practical applications, the accuracy and real-time performance of DT models depend on sensor technology and data transmission efficiency. As construction progresses and the environment changes, how DT models can efficiently adapt to and reflect the situation remains a major challenge. Furthermore, integrating safety rules and mechanisms into DT models to address the significant uncertainties in the construction process is also a key focus in the design of DT systems [41].

2.3. Research gaps and motivations

Although great achievements have been made in AI technologies and digital systems, several research gaps for HRIs collision warning in construction sites still remains.

Firstly, the collaborative analysis of multi-modal data for HRI collision warning is still limited. While current studies can achieve location tracking, pose estimation, and activity recognition of on-site objects, they fail to fully leverage multi-modal data from diverse sensors. This deficiency hinders a comprehensive understanding of on-site entities, particularly in addressing blind spots, which often results in workers failing to receive warnings about impending collisions. Without integrated multi-sensor data to realize comprehensive perception and address blind spots of construction sites, the accuracy and timeliness of collision warnings will be significantly compromised.

Secondly, most existing research has focused on conceptual investigations of DT, with insufficient exploration into the anti-collision system. A DT model capable of realizing closed-loop control encompassing perception, analysis, collision warning and control remains underdeveloped. Such a model is vital for dynamically responding to dangers, including sudden movements near blind spots, and ensuring that warning information is not only generated but also effectively transmitted to workers in real time.

Thirdly, there is a lack of robust digitalization systems specifically tailored for HRI collision warning in construction sites. In practical applications, the integration of DT with other technologies faces issues such as compatibility and data security. Barriers exist in data interaction and sharing between different technical platforms, restricting their wider and deeper application in smart construction. Additionally, current research on maintenance and update mechanisms for DT during long-term construction processes is inadequate. As construction progresses and environments change, DT models struggle to adapt and reflect actual conditions quickly and effectively, impacting their continuous functionality.

Therefore, this paper presents a digital twin-enabled anti-collision system designed to efficiently integrate the “perception, analysis, decision-making, and action” cycle for HRI. By addressing these gaps, especially for multi-modal data integration and closed-loop control, this system aims to significantly improve construction safety.

3. Digital twin-enabled anti-collision system

The proposed digital twin-enabled anti-collision system adopts the five-dimensional DT framework, which includes physical entity layer (PE), digital data layer (DD), virtual entity layer (VE), services layer (SS), and connections (CN). Functions of these five components are described in subsections, the framework of digital twin-enabled anti-collision system is shown in Figure 2.



Figure 2. The framework of digital twin-enabled anti-collision system.

3.1. Physical entity layer

The PE layer is the foundation of the DT system, which is responsible for the perception and control of on-site entities. PE consists of three parts: smart entities, data acquisition and transmission system, and smart construction site.

3.1.1. Smart entities

Smart entities encompass all entities in the construction site, such as excavators, trucks, workers, and facilities. Their real-time IDs, categories, statuses, positions, and working information will be synchronized to the DT system. Smart entities share and visualize their information within the DT system, as well as receiving and executing control instructions from the DT system.

Smart entities can be divided into two types: passive entities and active entities. Passive entities typically have a specific function and do not actively move, such as fences and warning signs found in construction sites. Key information of passive entities can be sensed and uploaded to the DT system for simulation, including their ID, categories, positions, and dimensions. Active entities encompass workers

and construction machines responsible for operations in the construction project. The position, pose, movement and activity of construction machines need to be accurately monitored because they are often potential causes of accidents. As vulnerable roles in HRIs, workers should also be precisely located and equipped with smart wearable devices such as smart helmets and smart vests to receive warnings from the DT system. Both passive and active entities can perceive and interact with each other in the DT system.

3.1.2. Data acquisition and transmission system

Data Acquisition and Transmission System (DATS) is deployed in construction site to collect on-site information from smart entities, as well as to facilitate the real-time transmission and synchronization of on-site information and control instructions between the PE and the DT system. DATS utilizes smart sensors such as cameras, IMU sensors, and RGB-D cameras for on-site information collection, enabling comprehensive perception of smart entities in construction sites. Besides, DATS also utilizes wireless communication technologies such as Bluetooth and Wi-Fi to achieve communication and interaction between different smart entities in the PE layer, as well as between the PE layer and different layers in the DT system. Which provides the foundation for achieving real-time visibility, synchronization, and control between the PE layer and the DT system.

3.1.3. Smart construction site

The smart construction site is the integration of smart entities and construction environment, which is responsible for modeling these smart on-site entities into an integrated system. The collected on-site information about construction machines, workers, and other entities will be considered as a group and uploaded to the DT system for analysis. Based on the perception and analysis of each on-site smart entity, the interactive relationships between each smart entity will be established to monitor the safety of HRIs in construction sites. Moreover, the smart construction site will also be synchronized within the DT system for visualization and further analysis. The automatic perception and remote monitoring in the smart construction sites can effectively prevent collisions and improve the construction safety.

3.2. Digital data layer

The DD layer plays the role of analysis in the DT system. It mainly consists of three parts: multi-modal physical data, multi-modal analysis model and virtual data. The multi-modal physical data is collected from PE layer in the construction sites. The multi-modal analysis model can accurately analyze multi-modal physical data to generate virtual data that can comprehensively describe the HRIs status. The analyzed virtual data is utilized to drive the virtual entity in the DT system, so as to achieve the information synchronization between digital space and physical space.

3.2.1. Multi-modal physical data

The multi-modal physical data is collected to realize the comprehensive perception of machines and workers in construction sites. RGB-D camera is utilized to capture both the 2D images and 3D depth images, which are respectively used to represent the absolute positions of on-site objects in 2D space and 3D space. They can also be fused. The RGB pixel values of 2D image are assigned to depth image

to generate the colored point cloud. IMU sensors are used to collect motion data, including acceleration, angular velocity, and angles, which can be used to express the motion trends of objects and predict their motion trajectory. The multi-modal data collected in construction sites is shown in Figure 3.

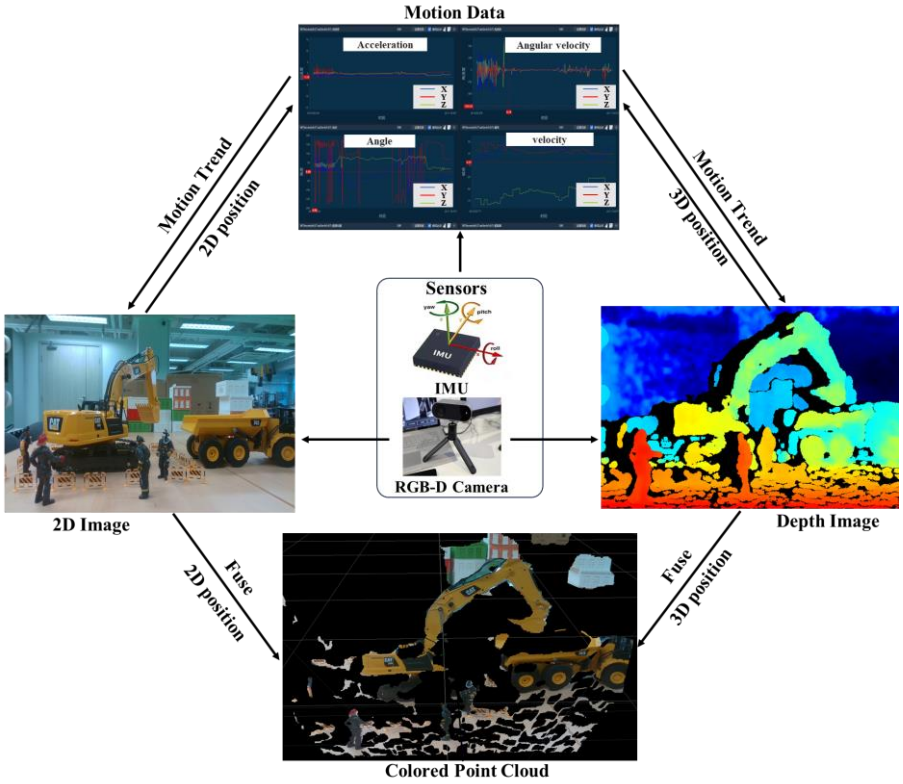
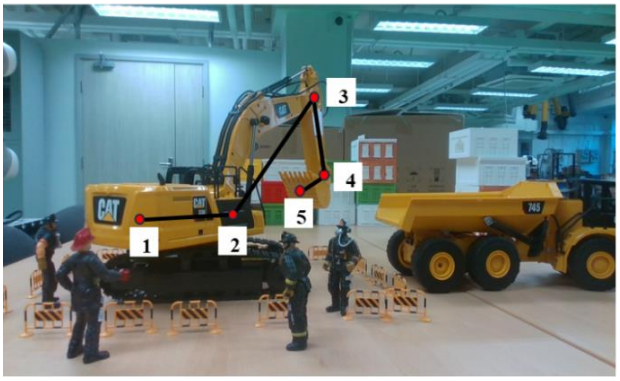


Figure 3. Multi-modal data collected in construction sites.

As one of the most important hazard sources in HRIs, it is crucial to accurately monitor the motion status of construction machines. In this context, the excavator has been chosen as the primary object for monitoring. Five key points are used to monitor the motion status of various parts of the excavator, including the body, boom, arm, and bucket, as shown in Figure 4. The IMU sensors are also attached to these parts to monitor detailed information on the motion dynamics of excavators. By combining the data from multiple IMU sensors and RGB-D camera, their motion status will be accurately monitored, and movement trends can be precisely predicted. This information can be utilized to prevent collisions and improve safety in the HRIs.




```
KEYPOINTs={
    1: 'body_end',
    2: 'cab_boom',
    3: 'boom_arm',
    4: 'arm_bucket',
    5: 'bucket_end',
}
```

Figure 4. Five key points of excavator.

3.2.2. Multi-modal analysis model

The DD layer employs AI model to analyze the multi-modal data collected from the PE layer. The multi-modal analysis model is designed to analyze collected 2D images, depth images, and motion data to obtain more valuable on-site information, such as categories, positions, key points, proximity, activities, and motion trends, the structure is shown in Figure 5.

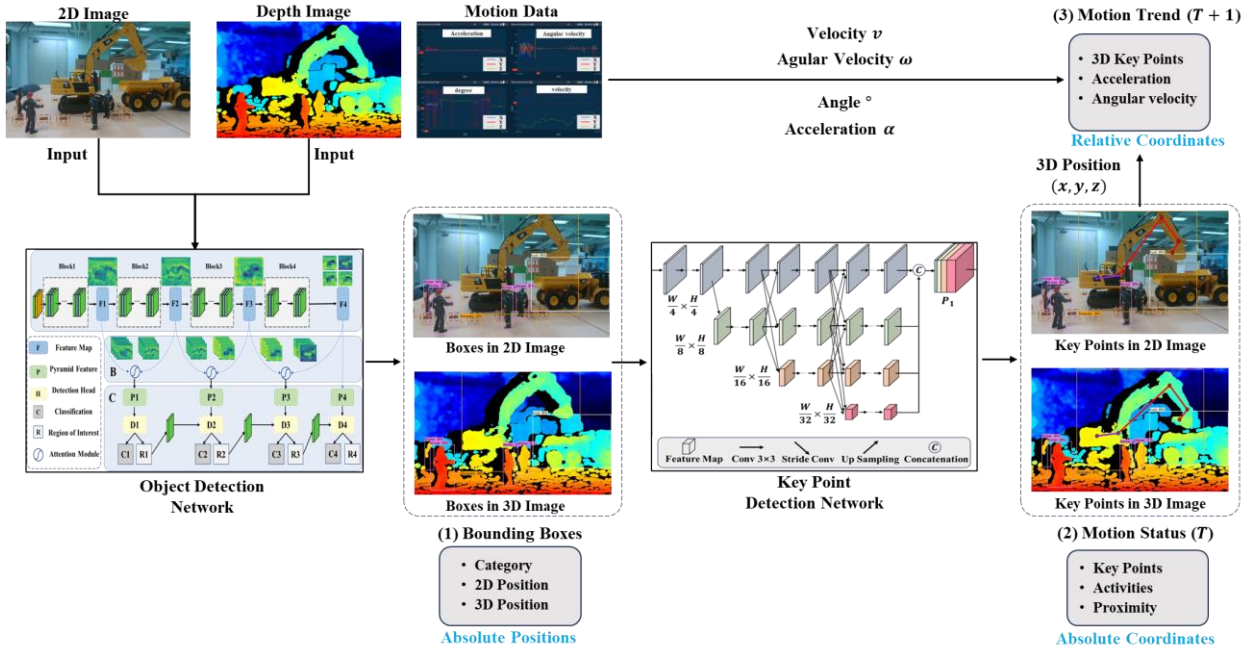


Figure 5. The structure of multi-modal analysis model.

Firstly, the object detection network is employed to predict bounding boxes and categories of construction machines and workers. To obtain their positions in both 2D and 3D spaces, the 2D image and depth image are fused to calculate the coordinates of these objects in the 3D space, their coordinates can be expressed as:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = D \begin{bmatrix} \frac{1}{f_x} & 0 & 0 \\ 0 & \frac{1}{f_y} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \quad (1)$$

Where x, y, z are the coordinates in the point cloud coordinate system, x' and y' are coordinates in the image coordinate in the image, and D represents the depth value.

Secondly, the key points detection network is utilized to detect five joints $\{E_1, E_2, E_3, E_4, E_5,\}$ of excavators. The coordinate (x, y, z) of these key points in 3D space can be used to represent their motion status and estimate their proximities to workers. Besides, key points can also be utilized to recognize the activity of construction machines.

Finally, the motion data collected from IMU sensors is utilized to augment the motion status and derive the motion trend of construction machines. This enables the prediction of the path and trajectory of construction machines in subsequent time periods. The degree °, acceleration α, velocity v, and

angular velocity ω of five joints $\{E_1, E_2, E_3, E_4, E_5,\}$ of excavators are used to represent their detailed motion status and motion trends.

3.2.3. Virtual data

The categories, bounding boxes, and key points detected from the multi-modal analysis model for on-site machines and workers are considered as virtual data. This data will be transferred to virtual entity layer and used to drive the motion of virtual models.

In addition, the DD layer also includes more types of data, such as algorithms, detection models, performance data from the SS layer, and historical performance data from the PE layer, simulation data and historical decisions from the VE layer. All the data in the DT system will be stored in the DD layer. By continuously interacting between physical entities and virtual entities, more data can be obtained to construct a more trustworthy DT system.

3.3. Virtual entity layer

The VE layer facilitates the integration of physical information and virtual information in the DT system. It consists of a geometry model, a physical model, a behavior model, and a rule model, which are responsible for decision-making.

3.3.1. Geometry model

In the DT system, geometry models are utilized to represent the geometric features (shape, size, position) and relationships (assembly) of physical entities. Detailed construction machines, facilities, and standard worker models will be generated in respective positions of construction sites. This provides geometry information of physical entities, allowing for the simulation, analysis, and prediction of HRIs in construction sites within the digital environment. Generally, AutoCAD, SolidWorks, 3D Max tools are employed for the 2D and 3D modeling.

3.3.2. Physical model

The physical model incorporates physical properties such as constraints, material, gravity, and joints to the geometry model. In this research, Unity is utilized to simulate physical behavior and scenes. Unity can set objects as rigid bodies and connect them using joints system, enabling the simulation of the physical structure, mechanical systems, motion, and constraints of construction machines. In addition, Unity also provides a robust collision detection system that can simulate contact, collisions, and corresponding events. Unity will be used to conduct process simulation, behavior simulation, evaluation, analysis, simulation, and other activities of the DT system.

3.3.3. Behavior model

In the DT system, behavior model refers to a model that simulates the behavior, operations, and function of physical system entities. Behavior models are constructed based on physical principles and algorithms. The behavior model (BM) utilizes accurate physical principles to predict the movement and changes of each entity from this second to the next, serving as feedback for decisions of the DT system.

The 3D positions of boxes and key points from virtual data are used to achieve precise synchronization between the digital entity and virtual entity at the current moment. For inactive entities such as workers, trucks, and facilities, their behavior is represented by bounding boxes. For active entities like excavators, the 3D key points are utilized to show their detailed motion status. Besides, the Rodriguez rotation method is employed to smoothly map the trajectory and pose of operations such as swinging and digging of excavators from virtual entities to physical entities. The pseudocode for calculation method is provided in Algorithm 1.

Algorithm1. The pseudocode of algorithm.

Algorithm 1 Calculate the operating Angle

Input: $v_{initial}$, v_{target}, $a_{initial}$.

Output: The angle of rotation of the machine relative to the parent component $a_{operating}$.

Define $v_{initial}$ and v_{target}.

$v_{initial} \leftarrow$ vector($[v_{initial_x}, v_{initial_y}, v_{initial_z}]$)

$v_{target} \leftarrow$ vector($[v_{target_x}, v_{target_y}, v_{target_z}]$)

Calculate the unit vector of $v_{initial}$

$v_{initial_{unit}} \leftarrow v_{initial} / \|v_{initial}\|$

Compute the rotation axis (cross product of $v_{initial}$ with the x-axis)

rotation_axis $\leftarrow v_{initial_{unit}} \times$ vector($[1, 0, 0]$)

rotation_axis_norm $\leftarrow \|$rotation_axis$\|$

if rotation_axis_norm $\neq 0$ **then**

    rotation_axis_unit $\leftarrow$ rotation_axis/rotation_axis_norm

else

    rotation_axis_unit $\leftarrow$ rotation_axis

end if

Calculate the rotation angle

angle $\leftarrow \arccos(\langle v_{initial_{unit}}, \text{vector}([1, 0, 0]) \rangle)$

Use the Rodrigues' rotation formula to calculate the rotation matrix

$$K \leftarrow \begin{bmatrix} 0 & -\text{rotation_axis_unit}[2] & \text{rotation_axis_unit}[1] \\ \text{rotation_axis_unit}[2] & 0 & -\text{rotation_axis_unit}[0] \\ -\text{rotation_axis_unit}[1] & \text{rotation_axis_unit}[0] & 0 \end{bmatrix}$$

rotation_matrix $\leftarrow I + \sin(\text{angle}) \cdot K + (1 - \cos(\text{angle})) \cdot K^2$

Apply the transformation to v_{target}

$v_{transformed} \leftarrow rotation_matrix \cdot v_{target}$

$a_{operating} \leftarrow \text{across}\left(\frac{[1,0] \cdot \left[v_{transformed_x}, v_{transformed_y}\right]}{\|[1,0]\| \cdot \left\|v_{transformed_x}, v_{transformed_y}\right\|}\right)$

$a_{operating} \leftarrow a_{operating} - a_{initial}$

Return $a_{operating}$

Except for the mapping from physical entities to virtual entities, data from IMU sensors is utilized to predict trajectory and path of excavators in the next moment, which can be expressed as follows:

$$\theta_{x,y,z}^{t+1} = \theta_{x,y,z}^{t} + \omega_{x,y,z} \times \Delta t \tag{2}$$

Where the θ_x^{t+1} means the angle in x, y, z axis at the $t+1$ moment, the ω_x indicates the angular velocity from IMU sensors, and $\Delta t = 1$.

3.3.4. Rule model

The Rule Model evaluates the collision risks of HRIs to make decisions using both TTC and SO strategies. As the "Code of Practice for Safe Use of Construction Machines" issued by the Labour Department of Hong Kong, a minimum clear distance of 600 mm should be maintained between working machines and fencing. Besides, construction sites also implement additional warning distances to ensure safety during operations. Therefore, a safe model divides the construction site into three zones: the dangerous zone, the warning zone, and the safe zone. Construction workers are only allowed to work in the safe zones, different levels collision warning will be triggered if any personnel enter the dangerous zone or the warning zone.

In this research, the excavator is used as the subject for the zone division of HRIs in construction site. During the operation, its movement mainly involves two parts: the body path and the trajectory of the arm, boom, and bucket. Both the moving body and working components (arm, boom, bucket) are considered in zone division. Notably, unlike conventional rule-based models that often restrict collision warning calculations to 2D planes, this research extends the spatial scope to a 3D workspace. To achieve this, the excavator 3D bounding box is first detected, which fully encompasses its body, arm, boom, and bucket in 3D space. Based on this 3D detection, dangerous, warning, and safe zones on construction sites are predicted, as illustrated in Figure 6. This 3D expansion enables more accurate and realistic zone delineation by accounting for vertical height differences (e.g., boom and bucket lifting/lowering)—critical for avoiding collisions in complex construction environments.

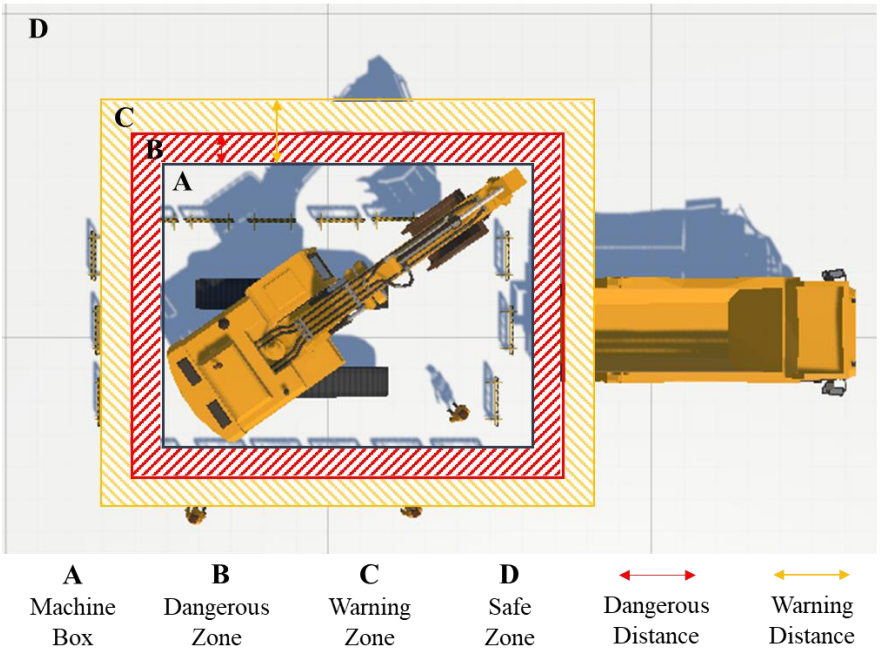


Figure 6. Zones divisions for HRIs in construction sites.

The danger distance is set to $d_{\text{danger}} = 600$ millimeters, while the warning distance is set to $d_{\text{warn}} = 1200$ millimeters. The regions maintained at a specific distance from the working construction

machines are considered as the dangerous zone and warning zone. The predictions of workers' situations at time $t + T_w$ are used to compared with the ground truth so as to assess the collision warnings methods. The t is current time, and T_w is warning time. The warning time T_w can be utilized to adjust the sensitivity of the anti-collision system.

The detected absolute positions $S_t^{(i)} = \left(x_t^{(i)}, y_t^{(i)}, z_t^{(i)}\right), i \in \{1,2,3,4,5\}$ of five key points at time t are used for zone predictions. To represent the warning zones and dangerous zones, horizontal plane of the construction sites is divided into $M \times N$ grids. These grids can be represented as $\Omega^{\text{grid}} = \{S(x_j, y_k)\}_{j=1,k=1}^{M,N}$. Besides, five key points are used to represent the motion status of machines, their matrix can be represented as $K_t = \left(S_t^{(1)}, S_t^{(2)}, S_t^{(3)}, S_t^{(4)}, S_t^{(5)}\right)$. The motion data acceleration vector $a^{(i)} = \left(a_{x,t}^{(i)}, a_{y,t}^{(i)}, a_{z,t}^{(i)}\right)$ and the velocity $v_t^{(i)} = \left(v_{x,t}^{(i)}, v_{y,t}^{(i)}, v_{z,t}^{(i)}\right)$ of the five key points can be utilized to calculate the motion status:

$$\mathrm{S}_{\mathrm{t}+\tau}^{(\mathrm{i})} = \mathrm{S}_{\mathrm{t}}^{(\mathrm{i})} + \mathrm{v}_{\mathrm{t}}\tau + \frac{\mathrm{a}_{\mathrm{t}}^{(\mathrm{i})}\tau^2}{2}, \tau \in [0, \mathrm{T_w}], \mathrm{i} \in \{1,2,3,4,5\}. \tag{3}$$

The warning matrix at $t + T_w$ is given by $\overline{K}_{t+T_w} = \left(S_{t+T_w}^{(1)}, S_{t+T_w}^{(2)}, S_{t+T_w}^{(3)}, S_{t+T_w}^{(4)}, S_{t+T_w}^{(5)}\right)$. Let the $S_t^{(2)}$ be the rotation center of the excavator body be the origin of coordinates, the working zones of machines can be defined directly by:

$$\mathrm{d}_{\mathrm{t}}^{\mathrm{zone}} \coloneqq \max_{\substack{\mathrm{i}\in\{1,3,4,5\} \\ 0\le\tau\le\mathrm{T_w}}} \left\| \mathrm{S}_{\mathrm{t}+\tau}^{(\mathrm{i})} - \mathrm{S}_{\mathrm{t}+\tau}^{(2)} \right\| \tag{4}$$

The point in the horizontal plane that is farthest from the rotation axis of the excavator must be located at the joints of the excavator. Therefore, the key points at the joints of the excavator as the criteria for determining the dangerous zones and warning zones, which can be defined by:

$$\Omega_{\mathrm{t}}^{\mathrm{danger}} = \Omega^{\mathrm{grid}} \cap \{(\mathrm{x},\mathrm{y}): \left\|\mathrm{x} - \mathrm{x}_{\mathrm{t}}^{(2)}\right\| \le \mathrm{d}_{\mathrm{t}}^{\mathrm{zone}} + \mathrm{d}_{\mathrm{danger}}, \left\|\mathrm{y} - \mathrm{y}_{\mathrm{t}}^{(2)}\right\| \le \mathrm{d}_{\mathrm{t}}^{\mathrm{zone}} + \mathrm{d}_{\mathrm{danger}}\} \tag{5}$$

$$\Omega_{\mathrm{t}}^{\mathrm{warn}} = \Omega^{\mathrm{grid}} \cap \{(\mathrm{x},\mathrm{y}): \left\|\mathrm{x} - \mathrm{x}_{\mathrm{t}}^{(2)}\right\| \le \mathrm{d}_{\mathrm{t}}^{\mathrm{zone}} + \mathrm{d}_{\mathrm{warn}}, \left\|\mathrm{y} - \mathrm{y}_{\mathrm{t}}^{(2)}\right\| \le \mathrm{d}_{\mathrm{t}}^{\mathrm{zone}} + \mathrm{d}_{\mathrm{warn}}\} \tag{6}$$

The d_{danger} and d_{warn}, represent the dangerous distance and warning distance in construction sites. They are set as related laws and rules. In this case, the workers are suggested to keep outside of the dangerous zones Ω_t^{danger}, and be careful in the warning zone Ω_t^{warn}.

3.4. Services layer

The SS layer provides is responsible for the execution in the DT system. It receives data, analysis, and decisions from the DT system and offers visualization services for all users. Site managers can monitor the detailed status of the construction site in the SS layer, so as to review historical decisions, adjust algorithms, and train models. The services contain supervision, zone prediction, collision warning, motion monitoring, and historical data management. These services can be provided to all users (clients, site manager, workers) through website. The layout of service website is shown in the Figure 7.

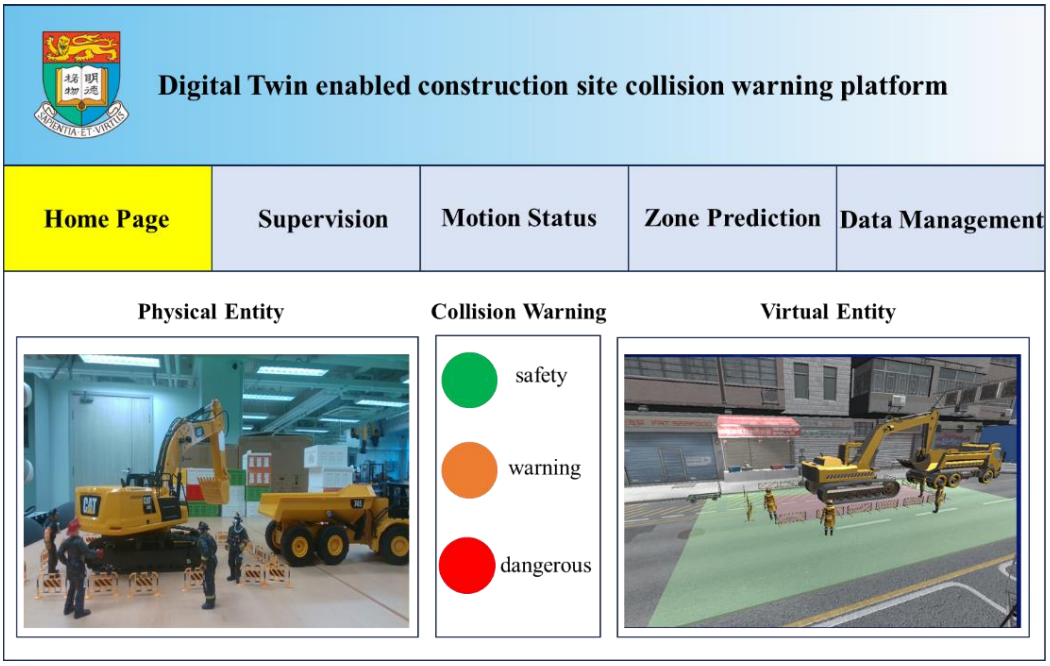


Figure 7. The layout of service website.

The supervision provides on-site information such as the categories, quantities, and positions of on-site objects. This can help site managers quickly understand the information of the construction site.

The zone prediction visually presents the dangerous zones, warning zones, and safe zones of the construction site, enabling all users to see the zones where workers are located. These zones can also be adjusted by users based on actual conditions. Additionally, collision warnings can be implemented by comparing the positions of workers with dangerous zones and warning zones in the future.

Motion monitoring provides detailed records of the motion status of construction machines, which includes absolute and relative positions such as acceleration, velocity, angle, and angular velocity. Not only site managers can have a detailed understanding of the motion status of construction machines, but operators can also have knowledge about the absolute position of machines in construction site as well as their relative positions to surrounding objects.

The historical data management handles various data from all layers, including models, algorithms, decisions, simulations, and detection results. Managers can analyze the motion status, simulation results and detection results of the DT system at any time. When errors are found in the simulation, modifications can be made to the geometry model and physical model of VE layer. When false collision warning happens, the behavior model and rule model should be adjusted. Additionally, the multi-modal analysis algorithms in the DD layer can also be optimized as detection results. Specific data can be selected to for the training of new detection model.

3.5. Connections

CNs are responsible for the interconnection and information sharing between the PE layer, DD layer, VE layer, and SS layer, as shown in Figure 8. Therefore, the closed-loop control of "perception, analysis, decision-making, execution" in the DT system can be achieved.

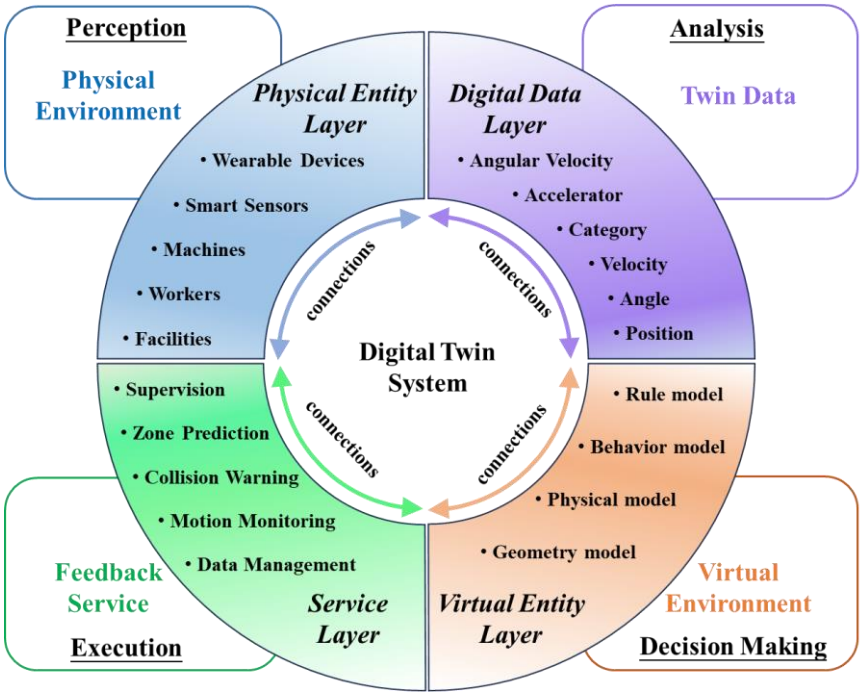


Figure 8. The connections for the DT system.

(1) Connections between the PE layer and other layers can enable accurate perception and information synchronization in the DT system. The static data and dynamic data from the PE layer will be used for data analysis, model driving, and decision execution. Moreover, the simulation results and decisions will be transmitted back to the PE layer for dynamic control, analysis, and evaluation.

(2) Connections between the DD layer to other layers support real-time storage and analysis in the DT system. The on-site data from the PE layer will be stored and analyzed in the DD layer. Besides, all data such as detection model, decisions and operation data will be continuously accumulated in the DD layer. These historical data stored in the DD layer can also be used to optimize the operations in the PE layer.

(3) Connections between the SS layer to other layers can provide visualization and execution for the DT system. The detailed on-site information from the PE layer, the VE layer and the DD layer will be visualized in the SS layer. Besides, the decisions and solutions will also be executed in the SS layer. The SS layer can analyze the performance of decisions and models, thereby selecting suitable solutions for optimizing the performance of the PE layer.

(4) Connections between the VE layer and other layers can enable the physical-virtual synchronization of the DT system. The static data and dynamic data from the PE layer and the DD layer will be utilized to drive the digital models in the VE layer. Simulation results and decisions from the VE layer will be sent back to the PE layer and the SS layer for visualization and dynamic control in the DT system.

4. Case study

4.1. Implementation of the digital twin-enabled anti-collision system

A case study is conducted to validate the effectiveness of the proposed digital twin-enabled anti-collision system. The implementation of the digital twin-enabled anti-collision system consists of physical space, digital space, and information synchronization, as shown in Figure 9.

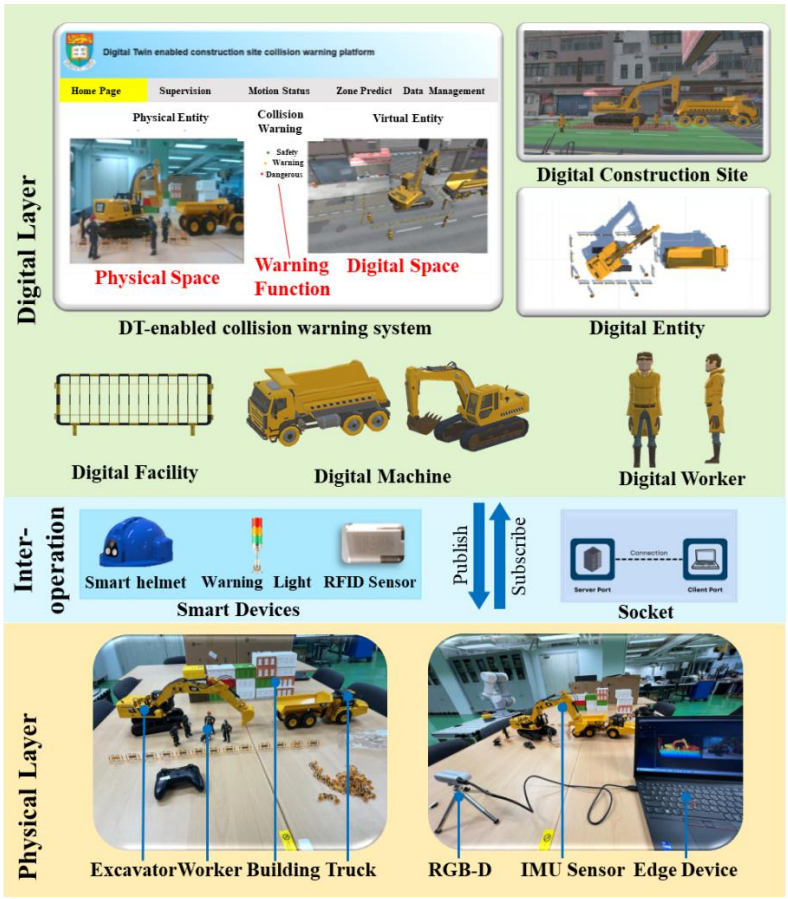


Figure 9. The implementation of digital twin-enabled anti-collision system.

In the physical space, a remote-controlled hydraulic excavator, remote-controlled articulated trucks, worker models, and construction facility models are used to simulate human-robot interactions (HRIs) on construction sites. The remote-controlled excavator, remote-controlled trucks, and construction facilities are scaled at 1:24 relative to real-world objects. The worker model is scaled at 1:20 relative to real-world objects, with the corresponding configuration listed in Table 1. 3D-printed buildings are used to simulate the construction environment. Additionally, smart helmets and tri-color warning lights are employed to receive collision warnings from the DT system. An RGB-D camera with a resolution of 1920 × 1080 is used to capture images and depth data, while IMU sensors are utilized to collect real-time motion data.

Table 1. Configuration of the DT-enabled anti-collision system.

Category	Length	Width	Height	Scale Ratio
Hydraulic Excavator	470 mm	155 mm	320 mm	1:24
Articulated Truck	430 mm	145 mm	150 mm	1:24
Construction Worker	25 mm	10 mm	90 mm	1:20
Construction Fence	55 mm	5 mm	25 mm	1:24

In the digital space, a Windows 11 system with an NVIDIA GTX 4060Ti is used for training and running the detection model, which is compiled using Python 3.11.7 and PyTorch 2.3.0. Real-time on-site

information—such as category, position, and motion status—is utilized to drive the operation of virtual entities. SolidWorks and 3D Max are employed to model virtual entities based on their CAD drawings. Additionally, Unity 3D is used to assign physical properties to these entities and establish their digital twin (DT) functions, which are then transferred to the DT system via WebGL. The interfaces of the DT system are custom-designed according to stakeholders' requirements.

To achieve information synchronization between the physical and virtual spaces, Socket and UDP protocols are adopted for information transmission and reception. Real-time on-site information processed and detected by Python is sent to Unity 3D via Socket. Unity 3D uses C# and UDP to receive this information and update it in the corresponding interfaces of the DT system. This on-site information, along with properties and models, is integrated using C# and then instantiated as virtual entities with N-dimensional information in Unity 3D, thereby enabling synchronization between the physical and virtual spaces.

This research collected a total of 30 videos with a total duration of approximately 20 minutes. These videos cover over 417 behaviors performed by excavators and trucks, including typical operating behaviors such as swinging, digging, moving, and lifting. All video frames are annotated with 2D bounding boxes for excavators, trucks, and workers to define their spatial positions in the RGB-D image. For excavators, five key points are further annotated to capture their motion, namely Body_End (rear end of the body), Cab_Boom (joint between cab and boom), Boom_Arm (joint between boom and arm), Arm_Bucket (joint between arm and bucket), and Bucket_End (tip of the bucket). Besides, the IMUs data collected from these five keypoints are used to capture the motion trend of excavators. Based on these annotations of boxes and motion data the behavior process, the operating zones of the excavator are labeled as 241 dangerous samples, 834 warning samples, and 593 safe samples. The images extracted from the videos are divided into training set, validation set, and testing set as 7:1:2.

4.2. Services of proposed digital twin-enabled anti-collision system

The services of the digital twin-enabled anti-collision system are also developed to provide visualization, supervision, motion status monitoring, zone prediction, and historical data management for all users, as shown in Figure 10.

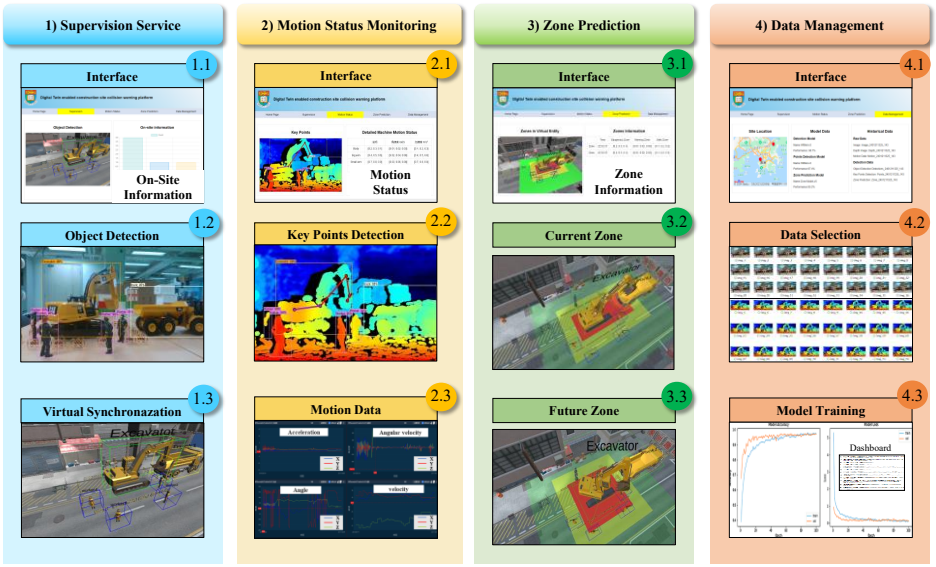


Figure 10. The services of the digital twin-enabled anti-collision system.

(1) Supervision service: Deep learning and computer vision methods are used to collect on-site information (e.g., the category and location of construction machines and workers). This information is also displayed on the interface for visualization. Additionally, corresponding 3D digital entities can be generated in the virtual space based on the collected real-time information. These digital entities synchronize with actual workers and equipment on the construction site, accurately reflecting their positions and states. Users can gain real-time insights into the actual conditions of the construction site and compare them with the digital entities. This real-time visual feedback and comparison help users manage the construction process more effectively, enabling them to promptly identify and resolve issues.

(2) Motion Status Monitoring: Computer vision and deep learning algorithms are used to detect key points of construction machines and obtain their 3D spatial coordinates in RGB-D images. Additionally, IMU devices are installed at these key points to monitor their motion data, such as acceleration, velocity, angular velocity, and angular acceleration. Using the current absolute position detected from RGB-D data and the motion trends collected by the IMU, the future positions of construction machines can be predicted. Both the current 3D coordinates and future 3D coordinates can be visualized in the DT system for all users, enabling better HRIs on construction sites.

(3) Zone Prediction: In the DT system, both virtual 3D digital entities and their corresponding zones can be visualized on the interface. Users can also monitor the 3D digital entities and zones from different perspectives. During the construction process, dangerous zones, warning zones, and safe zones are displayed in different colors on the construction site. When construction workers are detected in dangerous or warning zones, the system can automatically issue alerts of varying degrees. Furthermore, before actual execution, users can simulate the movement of construction machines to proactively identify potential hazards. If the simulation indicates a collision, users can identify the root causes and mitigate risks.

(4) Historical Data Management: All data in the DT system is stored and managed in the data management module, including raw data, detection data, model performance data, and error data. Users can analyze and explore various types of data in the DT system to discover potential correlations. By analyzing the stored data, users can select suitable data for training new models and integrate the trained models into the DT system, thereby enhancing the accuracy and reliability of modeling and simulation.

4.3. Detection and collision warning performance analysis

In this research, deep learning-based detection methods, including Faster RCNN [42], YOLOv5 [43], and HRNet [44], are used to evaluate detection performance. Faster RCNN utilizes ResNet50 as its backbone, while YOLOv5 employs DarkNet53 as its backbone. HRNet has three parallel feature extraction branches and a depth of 48 layers. Precision (P) and Recall (R) are used to evaluate detection performance, calculated as follows:

$$\text{Precision}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i} \tag{7}$$

$$\text{Recall}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \tag{8}$$

After achieving the detection of on-site objects, the HRNet with a top-down strategy is employed to detect the five key points of the excavator. The average deviations of each key point in the X, Y, and Z axes are calculated to evaluate the key point detection performance. The key point detection experiment is performed on physical models, with the detection results shown in Table 2.

Table 2. The average deviation between detected points and true points.

(mm)	X	Y	Z
Body_End	0.52	0.30	0.26
Cab_Boom	0.58	0.34	0.28
Boom_Arm	0.43	0.28	0.32
Arm_Bucket	0.30	0.24	0.18
Bucket_End	0.22	0.19	0.14

It can be observed from Table 2 that HRNet with a top-down strategy can achieve accurate key point detection. Among these five key points, Cab_Boom, Arm_Bucket, and Bucket_End are used to describe the operations of the boom, arm, and bucket. The average deviations for the arm and bucket are below 0.3 mm, and the average deviation for the boom is below 0.5 mm. Body_End and Cab_Boom are utilized to represent the movement of excavators. Their average deviations in the Y and Z axes are approximately 0.3 mm, while the average deviation in the X axis is about 0.5 mm. These results indicate that millimeter-level key point detection can be realized, and centimeter-level key point detection can also be achieved in real-world scenarios after ratio scaling. This enables precise key point detection of excavators and allows for comprehensive representation of their motion status.

In this research, the warning time is set as T_w = 3000 ms, the collision warning performance evaluation results are shown in Table 3.

Table 3. The collision warning performance evaluation results.

	Dangerous Zone	Warning Zone	Safe Zone
Number	241	834	593
Precision	96.36%	95.31%	97.36%
Recall	98.76%	97.36%	93.42%

The experimental results indicate that the proposed method in this paper achieves over 98% recall for dangerous zones and over 97% recall for warning zones. During the construction process, it can accurately predict dangerous zones and warning zones after a certain period of time, enabling early warnings to be issued to construction workers and helping avoid potential collisions in construction sites. Furthermore, actual testing has shown that the delay in collision warnings between the virtual space and the physical space is approximately 0.13 seconds, which can meet the requirements for collision warning and monitoring in on-site HRIs.

5. Conclusion and future work

In this research, a digital twin-enabled anti-collision system is designed to improve the safety for human-robot interactions in smart construction sites. This research has the following scientific contributions. (1) Multi-sensor fusion approach and multi-modal analysis model are developed to realize a comprehensive understanding of physical entities construction sites. (2) Geometry models, physical models, rule models and behavior models are designed to realize precise synchronization and collision warning in virtual space. (3) A 5D digital-twin enabled anti-collision system is proposed to achieve the closed-loop optimization of "perception, analysis, decision-making, execution" for HRIs in smart construction sites.

Although great progress has been achieved, there are still some limitations to be addressed in the future. When dangerous or warning zones are misjudged as safe ones, the system lacks a dynamic response mechanism (such as secondary verification, hierarchical emergency strategy) which may lead to hidden safety hazards being covered up. In addition, the "black box" nature of deep learning models will increase the uncontrollability of misjudgments, and the limitations of sensors may also affect the recognition accuracy. To solve the above challenges, future work will focus on the following directions:

(1) Improve the misjudgment response mechanism: Design a dynamic verification module. When the system judges a zone as safe, it will automatically trace back the sensor data and model reasoning process. If abnormal features are found, secondary monitoring will be triggered. At the same time, establish a multi-level response strategy, dynamically adjust the warning levels according to the misjudgment probability, and push manual review requests when necessary.

(2) Expand sensors and improve model interpretability: Introduce infrared thermal sensors, ultra-wideband sensors, and lidar to enhance the accuracy of entity recognition. Besides, attention mechanisms, model distillation and other technologies can also be studied to parse the decision logic of deep learning networks and reduce the "black box" risk.

(3) Strengthen system reliability: In response to the high safety requirements of the construction industry, optimize the real-time performance and robustness of the DT system, improve the anti-interference ability in complex scenarios through edge computing and cloud collaboration, and ensure the consistency of virtual-real mapping and the accuracy of early warning.

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Authors' contribution

Conceptualization, Shuxuan Zhao; methodology, Shuxuan Zhao; software, Chenglin Yu; validation, Shuxuan Zhao and Chenglin Yu; formal analysis, Shuxuan Zhao; investigation, Shuxuan Zhao; resources, Ray Y. Zhong; data curation, Shuxuan Zhao; Writing—original draft preparation, Shuxuan

Zhao and Chenglin Yu; writing—review and editing, Ray Y. Zhong; visualization, Chenglin Yu; supervision, Ray Y. Zhong; project administration, Ray Y. Zhong; funding acquisition, Ray Y. Zhong. All authors have read and agreed to the published version of the manuscript.

Conflicts of interests

The authors declare no conflict of interest.

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